

Student Performance Analysis: Descriptive and Predictive Analysis

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1. Data Cleaning and Preparation

To ensure data quality in datasets, data cleaning is an essential preprocessing step.

The original dataset StudentPerformanceFactors.csv has 6,607 rows with 20 columns, and was then transformed and cleaned with 6,606 rows and 12 columns to be used in our analysis.

1.1. Overview

The raw dataset contains 6,607 student records with 20 variables. Initial inspection revealed: 235 missing values across 3 columns, 1,463 invalid outliers in 2 columns, and several columns that were not directly relevant to our research question of what factors most influence student exam performance.

Table 1: Before vs After Cleaning Summary

Attribute	Before Cleaning	After Cleaning
Total Rows	6,607	6,606
Total Columns	20	12 (11 + 1 calculated)
Missing Values	235 (across 3 columns)	0
Invalid Hours (>24)	1,462 records	0 (replaced with median)
Invalid Exam Score (>100)	1 record	0 (replaced with median)
Duplicate Rows	1	0 (removed)
Derived Features	None	Score_Improvement

Table 2: Columns Removed with Justification

No.	Column Dropped	Reason for Removal	Data Type
1	Parental_Involvement	Not something the student can directly control, less relevant to this analysis	Categorical
2	Sleep_Hours	Indirect link to academic performance	Numerical
3	Motivation_Level	Based on self-report, making it subjective	Categorical
4	Internet_Access	92.4% “Yes”; doesn’t help distinguish outcomes	Categorical
5	Peer_Influence	External social factor, beyond core academic scope	Categorical
6	Physical_Activity	Lifestyle variable, indirect influence on performances	Numerical
7	Learning_Disabilities	89.5% “No”, highly imbalanced	Categorical
8	Parental_Education_Level	Parental attribute, and 90 missing values	Categorical
9	Distance_from_Home	Weak academic link, and 67 missing values	Categorical

1.2. Replacing Missing Values

After column selection, only 78 missing values remains, where each of them is located in the Teacher_Quality column (1.18% of records), as the other two columns with missing data (Parental_Education_Level and Distance_from_Home) had already been dropped. The 78 Teacher_Quality nulls were replaced with Mode, as the column is categorical variables. adding 78 values to this column increased the proportion from 60.1% to 60.6%.

1.3. Outliers Handling

Interval validation rules were implemented to detect logically impossible values. Hours_Studied was constrained to [0, 24] since a student cannot study more than 24 hours in a single day. Attendance, Previous_Scores, and Exam_Score were constrained to [0, 100] as percentage.

Table 4: Invalid Values and Replacements

Column	Valid Range	Invalid Count	Invalid Values	Replacement
Hours_Studied	[0, 24]	1,462	25–44	Median (20.0)
Exam_Score	[0, 100]	1	101	Median (67.0)
Attendance	[0, 100]	0	—	None needed
Previous_Scores	[0, 100]	0	—	None needed

A total of 1,462 records contained Hours_Studied values over 24, each were deemed invalid and then replaced with column median 20.0. One record contained an Exam_Score of 101, replaced with the column median of 67.0. The median was used because it is less affected by extreme values, ensuring the corrected outliers do not influence the replacement value. For Attendance and Previous_Scores, no invalid values were found.

1.4. Duplicate Data Handling

We identified one exact duplicate data, and removed it, reducing the dataset to 6,606 rows.

1.5. New Column: Score_Improvement

After cleaning, we create a new calculated column, where

$$Score_Improvement = Exam_Score - Previous_Scores.$$

This helps us see whether a student improved or declined compared to their previous academic standing. Positive values indicate improvement, negative values indicate decline, and zero indicates no change.

1.6. Summary

The final cleaned dataset contains 6,606 rows and 12 columns with zero missing values, zero invalid values, and zero duplicates, and will be used for the analysis in latter sections.

2. Descriptive Analysis

On this section, we will present the results of statistical measures and visualization that can better explain the dataset. Our goal here is to understand how each variable behaves on its own before drawing any deeper conclusions.

2.1. Statistical Measures and Interpretation

We computed several statistical measures for all six numerical variables, including measures of central tendency (mean, median, mode), dispersion (standard deviation, variance), and distribution shape (skewness).

Table 5: Descriptive Statistics for All Numerical Variables

Statistic	Hours Studied	Attendance	Previous Scores	Tutoring Sessions	Exam Score	Score Improvement
Mean	17.76	79.98	75.07	1.49	67.23	-7.84
Std Deviation	3.95	11.55	8.60	1.13	3.87	14.24
25th Percentile	16.00	70.00	69.00	1.00	65.00	-20.00
Median (50th)	18.00	80.00	75.00	1.00	67.00	-8.00
75th Percentile	20.00	90.00	82.00	2.00	69.00	4.00
Mode	20.00	60.00	73.00	0.00	67.00	-8.00
Variance	15.63	133.36	73.97	1.27	14.97	202.88
Skewness	-0.5831	-0.0007	0.0015	0.3503	2.1266	0.0401
IQR	4.00	20.00	13.00	1.00	4.00	24.00

2.1.1. Central Tendency (Mean, Median, Mode)

Starting with Exam Score, we obtained its mean 67.23, median 67.00, and mode 67.00. These numbers are not so far off each other, which we concurred that the distribution is fairly symmetric, with lots of students being around the 67 mark range, and no extreme stretch by outliers.

For Hours Studied, students put in in average (mean) 17.76 hours a week, with a median of 18.00. The two being so close together suggests a reasonably even spread, where most students are studying a moderate amount, and no clustering at an extreme.

Similarly, Attendance has a mean of 79.98% with a median of 80.00%, which means typical students tend to attend approximately 4 out of 5 classes.

The most visible information in these statistical measures, is the gap between Previous Scores (mean 75.07) and Exam Score (mean 67.23). Students were scoring noticeably higher on earlier assessments, shown also at the negative Score Improvement values.

As for Tutoring Sessions, the mean is only 1.49 with a median of 1.00, so most students are barely having any outside academic support, which is worth noting especially when we later ask whether tutoring actually makes a difference.

2.1.2. Skewness

Exam Scores are highly right-skewed (skewness = 2.13), suggesting that while most students achieved similar typical scores, a small number of students performed well and landed in the upper tail.

In contrast, Hours Studied is moderately left-skewed (-0.58), suggesting most students studied relatively more hours with only a few studying significantly less.

Attendance, Previous Scores, and Score Improvement show near-zero skewness, meaning approximately symmetric distributions.

Tutoring Sessions shows a slight right skew (0.35), meaning most students attended few sessions, while some attended more.

2.1.3. Summary

In summary, with the negative mean Score Improvement of -7.84 , we can conclude that on average, students scored nearly 8 points lower on the final exam compared to previous assessments. Since the median is also negative (-8.00), this is not driven by outliers but a genuine and widespread decline. This could indicate the final exam was more challenging, that student preparation was insufficient, or that previous scores included different assessment components.

2.2. Visualizations

From what we inferred on the previous section from the statistical measures, three charts were created to better visualize its results. For each chart, we will also present the relevance to the dataset, and insights.

2.2.1. Correlation Matrix Heatmap

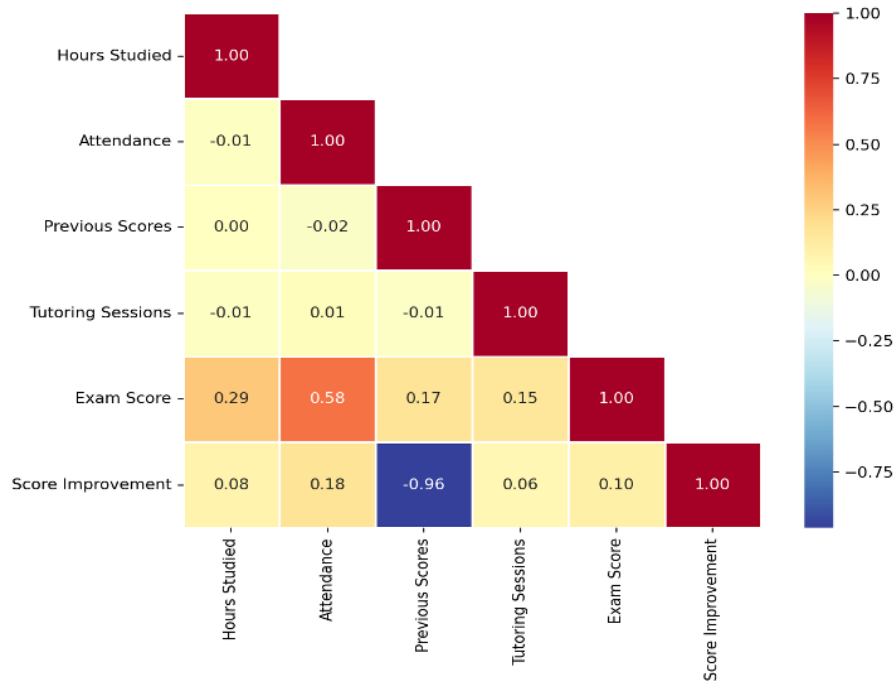


Figure 1 : Relations of each numerical variables.

Relevance: The heatmap helps us distinct which numerical variables have correlation with each other, and which of them are independent.

Insights: Hours Studied shows weak positive correlation with Exam Scores, indicating that there is a relation between the two factors, albeit weak. Tutoring Sessions also shows a very weak correlation with Exam Score. Attendance and Exam Scores on the other hand, shows a moderately positive correlation. Previous Scores and Score Improvement has a very strong negative correlation, a logical relation as higher previous scores give less room for improvement. In overall, this heatmap suggests that academic performance is not driven by just one factor but instead a mix of influences.

2.2.2. Hours Studied vs Exam Score

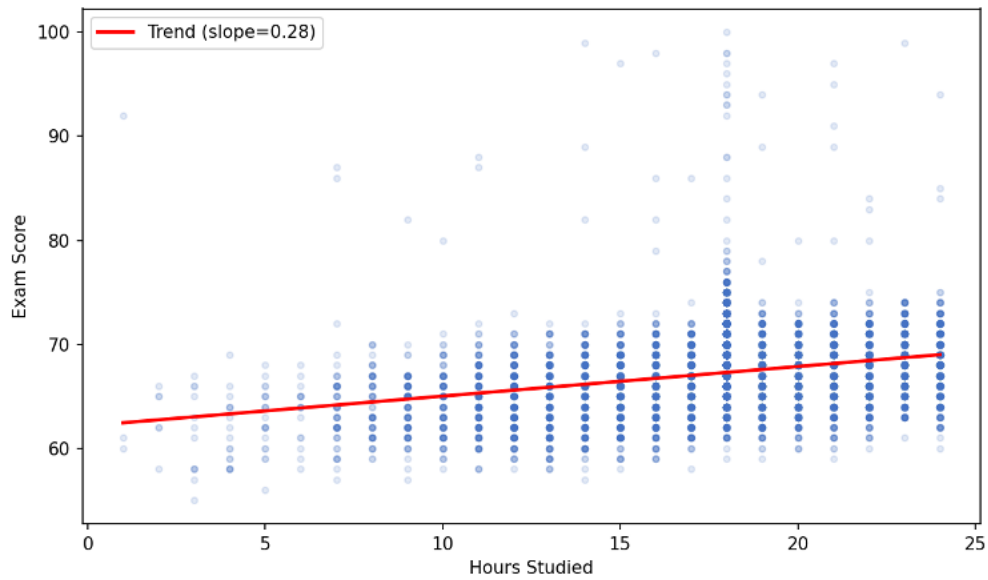


Figure 2: Hours_Studied vs Exam Score.

Relevance: Realistically speaking, study time is something students can directly control. Understanding whether more study hours lead to better outcomes is one of the questions this analysis would like to address.

Insights: In this scatter plot, the reference line has a moderate positive slope, aligned the weak positive correlation from the heatmap. This further suggests that factors beyond study time play a much larger role. Meaning, simply increasing study hours without addressing other factors will unlikely produce much improvements.

2.2.3. Attendance vs Exam Score by Gender

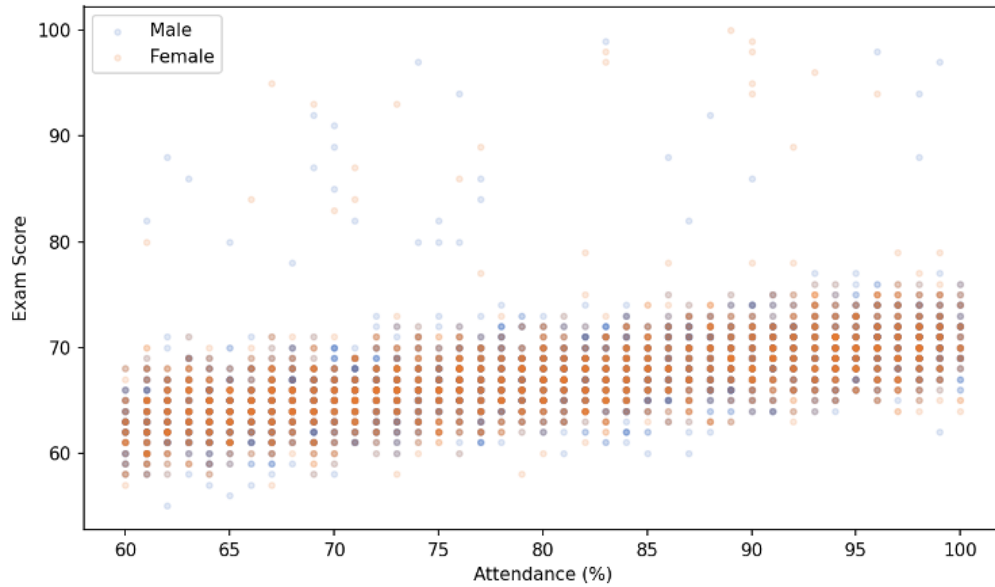


Figure 3: Attendance vs Exam Score for each gender.

Relevance: The scatter plot visualizes whether attendance is correlated with exam performance, and whether the relationship differs by gender.

Insights: Both genders have a same weak positive relationship between attendance and performance. Neither group shows a visibly different pattern. The density of points is similar for both colours at all attendance levels, showing that gender is not a meaningful aspect. We conclude that support on attendance would benefit all students equally, regardless of gender.

3. Data Dashboard & Justification

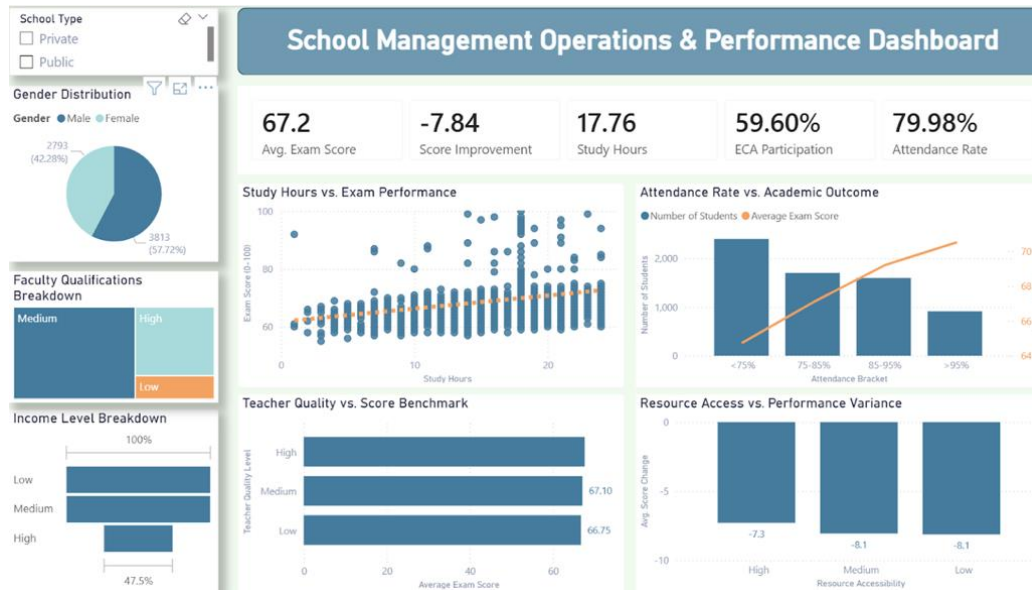


Figure 4: Data Dashboard.

Data Dashboard can be accessed [here](https://1drv.ms/u/c/C2F10646D44D955F/IQCPjvFYQIQoTK6_yLsma5gpAfkzoaQFsAcuU1dboHthWYs?e=Edrohn)

(https://1drv.ms/u/c/C2F10646D44D955F/IQCPjvFYQIQoTK6_yLsma5gpAfkzoaQFsAcuU1dboHthWYs?e=Edrohn)

All visualizations are interconnected, allowing the board to dynamically explore different students group and derive more targeted insights.

3.1. Target Audience

This dashboard is mainly designed for the Dean of Academic Affairs, as their responsibility is to look at things more closely on the academics of the university. By focusing on patterns in attendance, teaching quality, and student effort, the dashboard aims to help the Dean to improve academics more efficiently.

3.2. Charts & Justifications

3.2.1.KPI metrics

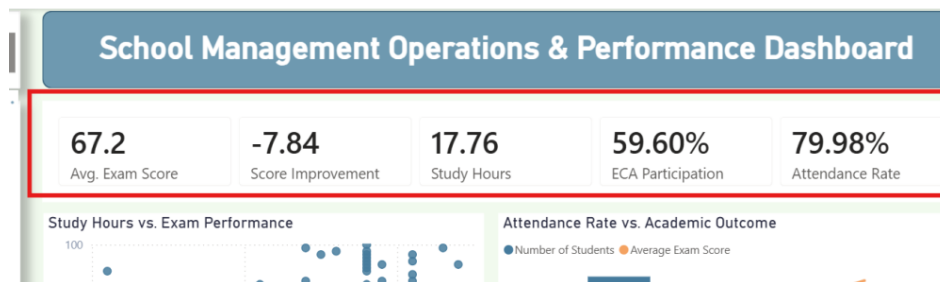
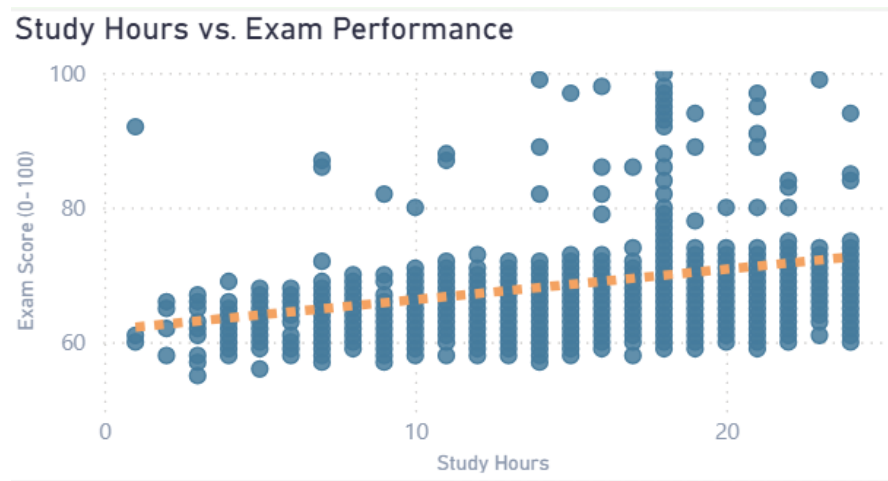


Figure 5: KPI Metrics

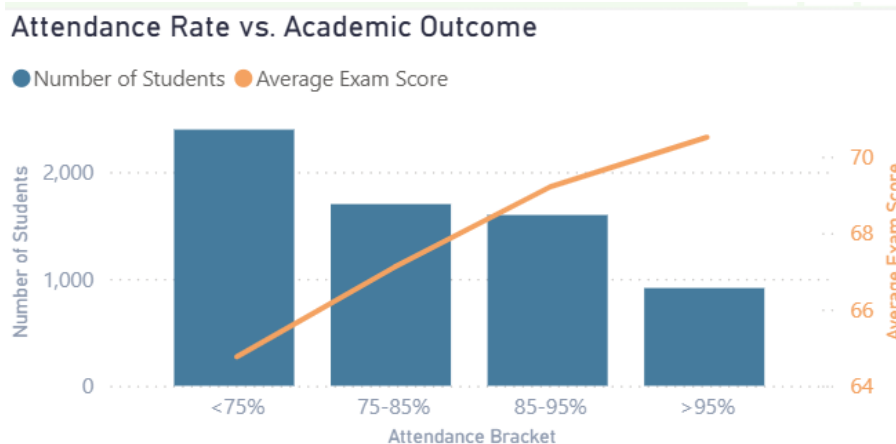
A set of 5 KPI cards was chosen to simply overall student performance. The key data highlighted are: average exam score (67.2), score improvement (-7.84), study hours (17.76), extracurricular activity (ECA) participation (59.60%), and attendance rate (79.98%). This provides the board with an instant overview to assess institutional health quickly.

3.2.2. Study Hours vs Exam Performance



A scatterplot was selected to visualise the relationship between study time and academic performance. This allows for a clear identification of any correlation between the two variables, helping to identify if increased study time is associated with higher scores. Such insights are valuable for policy-making decisions, enabling the university to implement targeted strategies that enhance student outcomes and overall institutional quality.

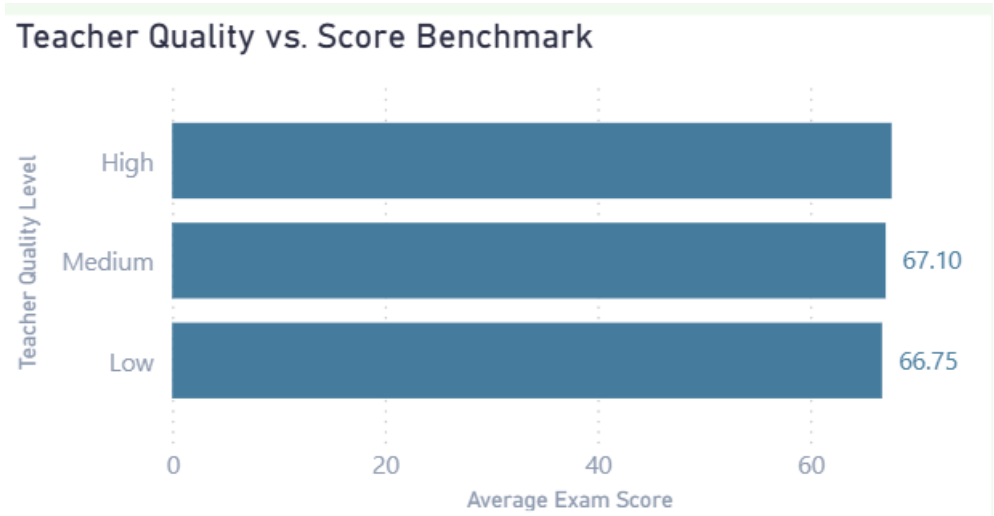
3.2.3. Attendance Rate vs Academic Outcome



A combined column and line chart was used to visualise attendance rate against academic outcomes, presenting both the number of students within each attendance bracket and their corresponding average scores. This dual-axis approach effectively captures both the

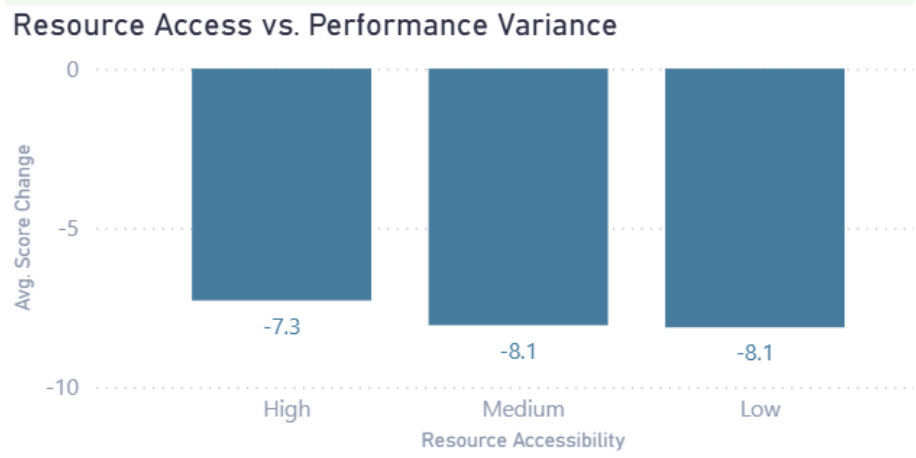
distribution of students and their academic performance, allowing for a more comprehensive interpretation of the data. By illustrating how average scores vary across different attendance levels, the chart highlights the impact of attendance on student achievement. These insights provide a strong evidence base for the university to justify and implement attendance-related policies or interventions aimed at improving overall academic performance.

3.2.4. Teacher Quality vs Score Benchmark



A bar chart is used to present the average academic scores across different levels of teacher quality, allowing for a clear comparison between categories. This highlights performance differences effectively, making it easier to assess whether higher teacher quality is associated with improved student outcomes. By visualizing this relationship, the chart supports data-driven evaluation of the return on investment (ROI) in teaching quality and provides a basis for decisions related to faculty development and training initiatives.

3.2.5. Resource Access vs Performance Variance



A bar chart is utilised to illustrate score changes across different levels of resource access. This visualisation identifies whether increased access to educational resources corresponds with measurable improvements in academic outcomes. By clearly displaying score differences, the chart allows the evaluation of resource allocation effectiveness and determines whether current investments are yielding meaningful benefits. This allows better planning of future budget distribution by ensuring that resources are directed towards areas where they can generate the most significant and consistent performance gains.

3.2.6. Demographic Charts

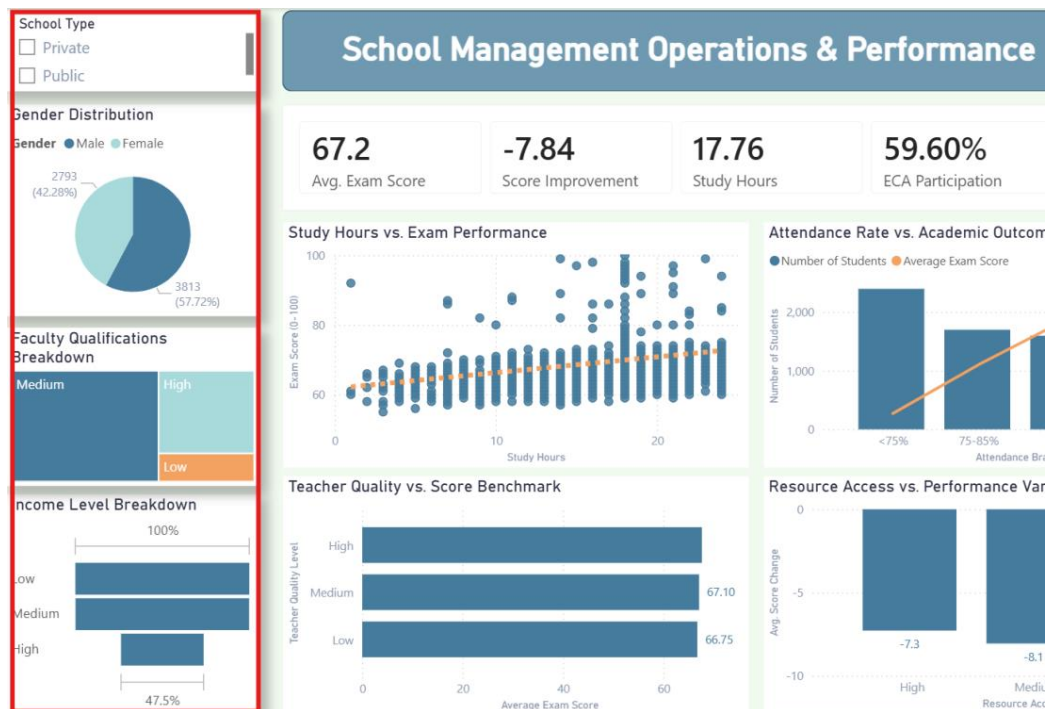


Figure 6: Demographic Charts.

A combination of pie chart, treemap, and bar chart is used to represent key student dynamic factors, such as previous school types, gender distribution, income levels, and faculty qualifications. These interactive visualisations provide contextual information that helps explain underlying patterns in student performance. By allowing users to filter and explore specific demographic segments, the dashboard enables deeper insight into how different groups are performing under varying conditions. This allows better identification of disparities in equity and accessibility.

4. Predictive Analysis

4.1. Potential Predictions

According to the Student Performance Factors dataset, the following predictions can be made:

- Student Exam Score Prediction: Student final exam scores to be predicted using the data of study hours, attendance, previous scores, and tutoring sessions.
- Student Risk Status Classification: The University needs to predict whether students fall into the “at-risk” category, such as failing an examination, and intervene in their academics early.

4.2. Suitable Predictive Techniques and its Variables

The following predictive analytics techniques are suggested for the prediction tasks above, and the specific variable(s) that will serve as input for each model are listed below.

4.2.1. Regression Analysis: Exam Score Prediction

Variable Type	Variables	Justification
Target Variable	Exam Score	As it is a variable that we would like to predict.
Predictor Variables	Hours Studied	To see how student’s efforts can affect their final exam scores.
	Attendance	As it represents the participation at class.
	Previous Scores	The strongest predictor for future performance.
	Tutoring Sessions	Indicating the use of academic support.
	Teacher Quality	The distribution of the test scores among different categories will show differences in analysis.
	Access to Resources	To see how resources affect students’ performance.
	Family Income	Socioeconomic background may impact academic performance.
	School Type	Different school environments may have varying risk levels.

Justification: With Exam_Score as a continuous numerical value, regression models like linear regression are suitable for this task. Because regression analysis aims to predict continuous outcomes based on input variables, this is not something achieved from a single numerical category. The selected predictor variables have been shown in descriptive analysis to have relationships with exam performance, hence they provide appropriate inputs for the proposed model.

4.2.2. Classification Models: Risk Status Prediction

Variable Type	Variables	Justification
Target Variable	Risk_Status (Binary: 0 = Exam Score $\geq$ 60, 1 = Exam Score < 60)	Redesigns prediction into task of binary classification which is consistent with deciding on interventions
Predictor Variables	Hours Studied	Students with very low study hours may be at higher risk
	Attendance	attendance is expected to correlate with higher risk
	Previous Scores	Students with low previous scores are more likely to be at risk
	Tutoring Sessions	High tutoring may indicate existing difficulties (reactive support)
	Teacher Quality	Teacher quality may influence student outcomes
	Access to Resources	Limited resources may increase risk probability
	Family Income	Socioeconomic background may impact academic performance
	School Type	Different school environments may have varying risk levels

Justification: Classification models such as Decision Trees and Logistic Regression are suitable because they produce interpretable results. Decision trees provide clear "yes/no" decision rules, while logistic regression outputs probability scores, allowing the institution to rank students by risk level. The selected predictor variables are all available before the final exam, enabling early identification.

4.3. Benefits of Predictive Analytics

Based on the predictive analysis above, the dataset can provide the following benefits to the institution:

- Early Identification of At-Risk Students: It helps identify the student who needs help in academics before the final examination and intervene early.
- Optimized Resource Allocation: It helps prioritized resource such as tutoring and counseling for high-risk students.
- Improved Student Success Rates: To help students improve learning performance through targeted intervention measures.
- Data-Driven Decision Making: Provide data support to university administrators to help develop more effective teaching policies.